



**RV College of
Engineering®**

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

2025 Scheme

CHEMISTRY CYCLE

Model Question Papers

RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
Department of Artificial Intelligence and Machine Learning
INTRODUCTION TO PYTHON PROGRAMMING
(2025 SCHEME)

Time: 03 Hours**Maximum Marks: 100****Instructions to Candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

Part A

Sl. No.		Questions	Marks	BTL	CO
1	1.1	Which data type is used to store a sequence of characters	1	1	1
	1.2	Compare / and // operators in Python.	1	2	1
	1.3	What will be the output of below Python code? str1= "information" print(str1[2:8])	1	3	1
	1.4	How do you create a tuple with only one element?	1	1	2
	1.5	Which method is used to return all keys from a dictionary?	1	1	2
	1.6	Write a function definition to add two numbers and return the result.	1	3	2
	1.7	Give an example of insert() and append() function for the list	1	2	2
	1.8	Write an example for Name Error Exception in Python.	1	3	3
	1.9	Which method in Pandas gives a statistical summary of numerical columns?	1	1	4
	1.10	Which method is used to identify missing values in a DataFrame?	1	1	4

Part B

UNIT - I					
2	a	Explain the different types of conditional statements and how they control the flow of a program. Provide examples to support your explanation.	9	2	2

	b	A software development trainee is building a small Python program that takes user information from the terminal, processes it, and stores it as text output. As part of the task: - The trainee needs to get input (such as name and age) from the terminal. - The trainee must convert the entered age (string input) into an integer to perform arithmetic operations. - Finally, the trainee should create and print a string message combining both values in a readable format. Illustrate how the trainee can implement the above requirements using suitable Python examples for: - Getting input from the terminal - Type casting - Creating strings	9	3	2
--	----------	---	---	---	---

UNIT - II					
3	a	What is a tuple? Explain immutability with an example. List any three built-in tuple methods with examples.	9	2	3
	b	In an engineering materials laboratory, the hardness test results of five metal samples are stored in a Python list: <pre>hardness = [45, 55, 60, 50, 48]</pre> Perform the following operations using appropriate list methods: - Change the hardness value of the second sample to 58. - Add a new sample hardness value 52 to the list. - Remove the lowest hardness value. - Organize the list in ascending order. - Display the final list after all operations. Also explain how the list data structure helps in solving this engineering data management problem.	9	2	2

UNIT - III					
4	a	Write Python code to perform the following operations on a dictionary: - Add a new key-value pair - Update the value of an existing key - Print all keys and values in the dictionary - Delete a key-value pair	9	3	2
	b	Differentiate between default arguments and keyword arguments? Explain with examples.	9	3	3
OR					
5	a	Explain the use of copy(), get(), has_key() and update() dictionary methods with syntax and examples.	9	2	2

	b	Differentiate between a user-defined function and a built-in function with an example and illustrate how function definition help in avoiding code repetition.	9	3	2
--	----------	--	---	---	---

UNIT – IV					
6	a	Differentiate between static methods and class methods with respect to their behaviour, syntax, and usage. Support your answer with suitable example for both methods.	9	3	2
	b	Explain the concept of encapsulation and its importance in Object-Oriented Programming. Illustrate how data hiding and controlled access are achieved in Python.	9	2	2
OR					
7	a	Illustrate how exception handling in Python helps manage runtime errors and ensures program stability. Support your answer with an example.	9	2	2
	b	A software company is developing a Vehicle Management System. The system needs to handle different types of vehicles such as Car and Bike, which share some common features like start_engine() and display_info(), but each has its own specific behaviour. Task: As a Python developer, you are asked to implement the following: - Create a base class Vehicle with a method display_info() that prints "This is a vehicle". - Derive two subclasses – Car and Bike – from the Vehicle class. - Override the display_info() method in both subclasses to print messages specific to each type of vehicle. - Demonstrate polymorphism by calling the display_info() method using a common reference for different objects.	9	2	2

UNIT – V					
8	a	How do you calculate the mean, median, and standard deviation of a NumPy array with a simple example.	9	3	3
	b	Explain how to customize plot titles, axis labels, and legends.	9	2	2
OR					
9	a	Create and demonstrate Data Frame from a dictionary, list of lists, or a NumPy array with simple examples.	9	3	3
	b	How do you create a NumPy array from a Python list? Provide an example for a 1D and a 2D array.	9	2	2

Signature of the Scrutinizer

Name:

Signature of the BoE Chairperson

Name:



RV College of Engineering®
Mysore Road, RV Vidyaniketan P
Bengaluru - 560059, Karnataka, In

CM211IA

USN

--	--	--	--	--	--	--	--	--	--

RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU, Belagavi)
I Semester B. E. Regular/Supplementary Examinations January-2026
MODEL QUESTION PAPER
Course: CHEMISTRY OF SMART MATERIALS AND DEVICES

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

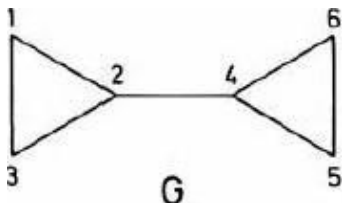
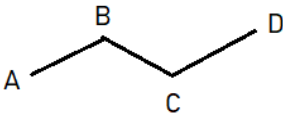
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. Question number 2 and 3 are compulsory. Choose any one full question from 4 or 5, 6 or 7 and 8 or 9.

PART-A

Question No.	Questions	M	CO	BT
1.1	Identify the key property that makes PMMA suitable for dental applications.	1	1	3
1.2	Accumulation of microplastics in the environment is difficult to remediate. Justify.	1	4	4
1.3	Mention any one advantage of OLEDs over traditional LEDs	1	2	4
1.4	What is the function of a capacitor in memory devices?	1	1	1
1.5	List any one structural or performance advantage of supercapacitors over conventional capacitors	1	1	1
1.6	EDLCs are better than conventional capacitors in terms of energy density. Justify.	1	2	5
1.7	Identify the electrodes used in potentiometry.	1	1	4
1.8	Compare single-walled and multi-walled CNTs in terms of structure.	1	2	4
1.9	How QSAR contributes to identifying potential drug candidates in pharmaceutical research	1	3	4
1.10	Which matrix shows the number of edges between atoms?	1	3	5

PART – B

Question No.	Questions.	M	CO	BT
2a	Construct the life cycle assessment of Polyethylene bag. Recommend an alternative product for the Polyethylene bag.	8	2	4
2b	Outline the synthesis of PMMA and discuss its properties. List any two biomedical applications.	6	1	2
2c	Interpret the effects of lead and mercury on human health.	4	4	2
3a	Illustrate the classification of memory devices.	8	1	3
3b	Analyse the differences in structure and function between liquid crystals, LEDs, OLEDs used in display technologies.	6	3	4
3c	Mention any two environmental benefits and limitations of green computing.	4	4	1
4a	Compare the electrochemical mechanisms and performance between Lithium Iron Phosphate and lithium-air batteries in terms of energy density, cycle life and safety.	8	3	4
4b	Evaluate the working principles in EDLCs and hybrid supercapacitors	6	1	
4c	List out any two advantages and limitations of inorganic solar cells over organic photovoltaics	4	2	4
	OR			
5a	Explain the construction and working of Quantum Dot Sensitized Solar Cells (QDSSCs).List any two applications.	8	3	2
5b	Apply your knowledge of battery management systems (BMS) to design a basic framework for monitoring and controlling battery packs in e-mobility.	6	1	3
5c	Outline the importance of light absorption and charge separation in photocatalytic water splitting.	4	2	2
6a	Explain the working principle and key components of conductometry. Explain how a calibration graph and conductivity formula are used to calculate ionic concentration in case of strong base <i>vs</i> mixture of acids.	8	4	2
6b	Apply your understanding of polymer chemistry to explain the synthesis of Polyaniline. List any two multifunctional applications.	6	2	3
6c	Illustrate the key steps involved in the synthesis of graphene by modified Hummer's method.	4	2	3
	OR			
7a	Describe the principle of flame photometry in analysing alkali metals. Explain the function of its main components and illustrate how a calibration	8	3	2

	graph is used to determine metal ion concentrations.			
7b	Analyse the role of electrochemical sensors in biomedical diagnostics, focusing on glucose with detection mechanisms.	6	2	4
7c	Discuss any two practical uses of Yttrium Barium Copper Oxide (YBCO) in modern superconducting technologies.	4	2	2
8a	Construct and interpret vertex adjacency and distance matrices for the following structure "G". 	8	3	3
8b	Apply the concept of topological indices to calculate the Wiener index for a simple hydrocarbon molecule butane. (CH ₃ -CH ₂ -CH ₂ -CH ₃)	6	3	3
8c	Mention any two differences between electrostatic interactions and hydrogen bonding in terms molecular orientation.	4	1	1
	OR			
9a	Apply QSAR principles to outline the steps involved in building a predictive model for toxicity screening.	8	1	3
9b	Mention the first and second Zagreb indices (M1 and M2) for the following molecule. 	6	3	1
9c	Calculate the Platt number for the below mentioned structure. 	4	3	1

Signature of Scrutinizer

Name: Dr. Raviraj Kusanur

Signature of Chairman

Name: Dr. Mahesh R

--	--	--	--	--	--	--	--	--	--

RV COLLEGE OF ENGINEERING®
An Autonomous Institution Affiliated to VTU
I Semester B. E. January 2026 Examinations
Computer Science Engineering
Fundamentals of Programming using C

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

PART-A			M
1	1.1	C programs are converted into machine language with the help of _____.	1
	1.2	Write the output of the following program: #include<stdio.h> main() { printf(“%d”,’a’); }	1
	1.3	_____ operator returns the number of bytes occupied by the operand	1
	1.4	The memory address of the first element of an array is called _____.	1
	1.5	The statement arr[3] = 10; initializes the _____ element of the array with value 10.	1
	1.6	What is the significance of the instruction strcpy (str1,str2)?	1
	1.7	Give two difference between Recursion and Iteration.	2
	1.8	The parameters passed to function are called _____ parameters.	1
	1.9	How will you define the symbolic constant to represent the π value in the C programming language?	1

PART B (Maximum subdivisions is limited to 3 in each questions)

UNIT I			
2	a	What is a computer? With a neat diagram explain the components of a computer system.	18
	b	With an example, explain the structure of C program.	
	c	What is the use of writing an algorithm? Explain the control structures which can be employed by algorithms with example for each.	
UNIT II			
3	a	Write a short note on basic data types that the C language supports.	18
	b	Differentiate between typecasting and type conversion	
	c	Write a program to count number of vowels and consonants	
UNIT III			
4	a	What do you understand by identifiers and keywords and explain its rules	18
	b	Write a program to convert degrees Fahrenheit into degree Celsius.	
	c	Write a program to check given number is palindrome	

OR

--

5	a	Explain the terms variables and constants. How many type of variables are supported by C?	18
	b	Write a program to calculate salary of an employee, given his basic pay (to be entered by the user), HRA=10% of the basic pay, TA=5% of basic pay. Define HRA and TA as constants and use them to calculate the salary of the employee.	
	c	Write a program to prepare a grocery bill. For that enter the name of the items purchased, quantity in which it is purchased, and its price per unit. Then display the bill in the following format.	

UNIT IV

6	a	Differentiate else-if ladder from switch-case. Write a program to read the numbers n1 and n2 and perform arithmetic operations according to input of user choice, for example if user enters choice value 1, then program should perform addition of n1 and n2 and so on.	18
	b	Illustrate the concept of nested if-else with a simple example.	
	c	Write a program that reads two matrices A and B, and compute and print the product matrix of A and B.	

OR

7	a	Write a program to compute and print the electricity bill as per the rates given in the following table according to the units consumed. <table><tr><td>Units</td><td>Rate(Rs)</td></tr><tr><td>00 and above</td><td>5.50/unit + 20</td></tr><tr><td>200-500</td><td>3.50/unit + 30</td></tr><tr><td>100-200</td><td>2.50/unit + 40</td></tr><tr><td>Less than 100</td><td>1.50/unit + 50</td></tr></table>	Units	Rate(Rs)	00 and above	5.50/unit + 20	200-500	3.50/unit + 30	100-200	2.50/unit + 40	Less than 100	1.50/unit + 50	18
Units	Rate(Rs)												
00 and above	5.50/unit + 20												
200-500	3.50/unit + 30												
100-200	2.50/unit + 40												
Less than 100	1.50/unit + 50												
	b	Write a program that reads an array of 100 integers. Then display all the pairs of elements in the array whose sum is 50.											
	c	Compare the following code snippets and mention both are finite loops or infinite loops or both are different loops and add a note on the logic of any one code snippet. <table><tr><td><pre>#include<stdio.h> int main(){ int i,j; sectionA: for (i=0,j=5;i<5,j>=0;i++,j--) { printf("%d %d",j,i); goto sectionA; } }</pre></td><td><pre>#include<stdio.h> int main(){ int i=0,j=5; for (;;) { printf("d %d",j,i); i++; j--; } }</pre></td></tr></table>	<pre>#include<stdio.h> int main(){ int i,j; sectionA: for (i=0,j=5;i<5,j>=0;i++,j--) { printf("%d %d",j,i); goto sectionA; } }</pre>	<pre>#include<stdio.h> int main(){ int i=0,j=5; for (;;) { printf("d %d",j,i); i++; j--; } }</pre>									
<pre>#include<stdio.h> int main(){ int i,j; sectionA: for (i=0,j=5;i<5,j>=0;i++,j--) { printf("%d %d",j,i); goto sectionA; } }</pre>	<pre>#include<stdio.h> int main(){ int i=0,j=5; for (;;) { printf("d %d",j,i); i++; j--; } }</pre>												

UNIT V

8	a	Using functions, write a program to generate prime numbers between given numbers.	18
	b	Write a C program to add two complex numbers using structures.	
	c	Define pointers. Explain the use of & and * operators in pointers with suitable examples.	

OR

9	a	Explain the concept of structures within structures with the help of an example.	18
	b	With the help of pointers, write a C program to add two numbers.	
	c	Explain the concept of pass by value and pass by reference with example.	

Q.P.Code

Course Code: CS113ATA

Course Title: Fundamentals of programming using C

PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	1	1	1	1	1	1	2	1	1	
COs	1	2	1	1	1	2	2	2	2	
Q No	1.11	1.12	1.13	1.14	1.15	1.16	1.17	1.18	1.19	1.20
B T										
COs										

PART-B

Question No		B T Levels	Cos addressed		Question No		BT Levels	Cos addressed
2	a	1	1		3	a	2	2
	b	1	2			b	2	2
	c	2	2			c	2	2
	d					d		
4	a	2	3		5	a	2	3
	b	3	3			b	3	3
	c	3	3			c	3	3
	d					d		
6	a	2	3		7	a	2	3
	b	3	4			b	3	4
	c	3	4			c	3	4
	d					d		
8	a	3	3		9	a	3	3
	b	4	4			b	4	4
	c	2	3			c	2	3
	d					d		
10	a				11	a		
	b					b		
	c					c		
	d					d		

Signature of Scrutinizer:
Name:

Signature of Chairman
Name:

USN

--	--	--	--	--	--	--	--	--	--

RV COLLEGE OF ENGINEERING®
An Autonomous Institution Affiliated to VTU
I Semester B. E. January 2026 Examinations
Computer Science Engineering
INTRODUCTION TO WEB PROGRAMMING

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

		PART-A	M	B T	CO
1	1.1	The _____ is a global network of interconnected computers that communicate using the TCP/IP protocol.	1	1	1
	1.2	The first line of an HTTP request message includes.....	1	2	1
	1.3	Differentiate between a web browser and a web server.	1	1	2
	1.4	The attribute used to specify alternative text for an image is _____.	1	2	1
	1.5	The general syntax of a CSS rule is selector { property: value; } where _____ specifies the element to be styled.	1	2	2
	1.6	In case of conflicting rules, the style with the _____ specificity takes precedence.	1	1	2
	1.7	What is meant by object orientation in JavaScript?	1	1	1
	1.8	How do you access the elements of an array?	1	2	1
	1.9	Write a regular expression to check if a string contains the word “web”.	1	3	2
	1.1 0	A database is a collection of _____ data organized for easy access and management	1	1	1

PART B (Maximum subdivisions is limited to 3 in each questions)

UNIT I			
2	a	Define URL and explain its structure with an example	06
	b	What is MIME? Explain its role and provide examples.	06
	c	Explain the HTTP request–response cycle with suitable examples.	06

UNIT II			
3	a	Explain the basic syntax rules of XHTML with suitable examples.	06
	b	Explain how images and hyperlinks are inserted in XHTML with examples.	06
	c	Create an XHTML table for a College Course Information Page that displays details of the courses offered by a department. Use proper XHTML syntax (lowercase tags, closed elements, quotes for attributes).The table should have the following specifications: • The table should have a caption: “Department of Computer Science – Course Details” • It must have three columns: 1. Course Code 2. Course Name	06

		3. Duration (in years) • Include four rows of course information. • The table should have a border, and header cells must be bold and centered.	
--	--	--	--

UNIT III			
4	a	Explain CSS. What are its advantages over traditional HTML styling?	06
	b	Write short notes on <code></code> and <code><div></code> tags in CSS.	04
	c	The college website header uses a background image <code>header_bg.jpg</code> . It should: • Repeat horizontally only. • Be fixed when the page scrolls. • The background color should be <code>#f0f8ff</code> if the image doesn't load. Write the CSS code to implement these requirements and briefly explain each property used.	08

OR

5	a	Explain the different levels of style sheets in CSS with examples.	06
	b	What is conflict resolution in CSS? Explain with example.	06
	c	A student designs a <code><div></code> container with the following properties: • Content width: 200px • Padding: 10px • Border: 5px • Margin: 15px Answer the Questions below: 1. Calculate the total space occupied by the element on the page. 2. Explain each part of the box model (content, padding, border, margin). 3. If <code>box-sizing: border-box</code> were used, how would the total width change?	06

UNIT IV			
6	a	Explain the main features and advantages of JavaScript as a web programming language.	06
	b	Explain the different ways to display output in JavaScript.	06
	c	Write JavaScript code to display marks of five students stored in an array and calculate the average mark. Discuss the importance of loops and arrays in repetitive computations in JavaScript.	06

OR

7	a	Explain how JavaScript supports object-oriented programming (OOP) concepts.	06
	b	Compare and contrast screen output functions — <code>alert()</code> , <code>confirm()</code> , and <code>prompt()</code> with examples.	06
	c	Create an interactive web page that contains a button labeled “Show Date”. When the user clicks the button, a JavaScript function should display the current date and time in a <code><p></code> tag.	06

UNIT V			
8	a	Describe the relational database model.	06
	b	Compare and contrast relational databases with NoSQL databases.	04

	c	Describe <code>SELECT</code> , <code>INSERT</code> , <code>UPDATE</code> , and <code>DELETE</code> with short example queries.	08
OR			
9	a	Discuss ACID properties in relational databases.	04
	b	Explain <code>COUNT</code> , <code>SUM</code> , <code>AVG</code> , <code>MIN</code> , <code>MAX</code> and grouping using <code>GROUP BY</code> and <code>HAVING</code> with examples.	14

Q.P.Code

Course Code: CS115AIB

Course Title Introduction to Web Programming

PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	1	2	1	2	2	1	1	2	3	1
COs	1	1	2	1	2	2	1	1	2	1
Q No	1.11	1.12	1.13	1.14	1.15	1.16	1.17	1.18	1.19	1.20
B T										
COs										

PART-B

Question No		B T Levels	Cos addressed		Question No		BT Levels	Cos addressed
2	a	2	1		3	a	2	3
	b	2	2			b	3	3
	c	3	3			c	4	4
	d					d		
4	a	2	3		5	a	2	3
	b	2	2			b	2	2
	c	4	4			c	4	4
	d					d		
6	a	2	3		7	a	2	3
	b	2	2			b	2	2
	c	4	4			c	4	4
	d					d		
8	a	2	3		9	a	2	3
	b	2	2			b	2	2
	c	4	3			c	4	3
	d					d		
10	a				11	a		
	b					b		
	c					c		
	d					d		

Signature of Scrutinizer:
Name:

Signature of Chairman
Name:

COURSE CODE- CV113ATB
USN-

RV COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

I Semester BE January -2026 Examinations

DEPARTMENT OF CIVIL ENGINEERING

COURSE- ELEMENTS OF CIVIL ENGINEERING

Time-03 Hours

Maximum Marks-100

Instructions to Candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only
2. Answer FIVE full questions from Part B. In Part B question number 2 & 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

PART A

Q No	Questions	Marks
1	1.1 Define Plinth area.	1
	1.2 Name the national green building rating system used in India.	1
	1.3 List various alternative materials used in civil engineering	2
	1.4 Define potable and palatable water.	2
	1.5 Differentiate between shallow and deep foundations	2
	1.6 List the applications of RS and GIS in civil engineering	2

PART B

UNIT-I			
2	a	Describe briefly the scope and significance of Civil Engineering in Structural Engineering	4
	b	Justify the need for multidisciplinary knowledge in Civil Engineering practice.	8
	c	Discuss the role of Civil Engineers in achieving sustainable development goals	6

UNIT-II			
3	a	Explain briefly the engineering properties of bricks.	4
	b	Explain the classification and functional importance of beams and columns in building construction	6
	c	Explain the concept of green buildings and discuss their need in sustainable development	8

UNIT-III			
4	a	Illustrate with a neat diagram the stages involved in a typical water treatment process and describe the functions of each stage	10
	b	Explain how rainwater harvesting systems help reduce dependency on surface water sources and discuss their role in achieving sustainable urban water resource management	8
5	a	Bengaluru city, with a population exceeding 1.4 crore, is presently facing acute challenges in managing municipal solid waste due to rapid urbanization and population growth. The increasing quantity of waste has overburdened collection, segregation, and disposal systems, leading to environmental pollution and health hazards. For the above context, - Identify and classify the major types of solid waste generated in Bengaluru city. - Suggest sustainable practices and innovative solutions that can be adopted to minimize waste generation and enhance recycling efficiency?	10
	b	Discuss briefly the importance of dams, canals, and aqueducts in water resource management	8

UNIT-IV			
6	a	Explain the classification of roads in India	8
	b	Explain the role of public transport systems in reducing traffic congestion and discuss the importance of intermodal connectivity in achieving efficient urban mobility	10
7	a	Explain briefly explain the classification of tunnels	8
	b	Discuss the importance of road safety awareness programs and describe how modern RS and GIS tools assist authorities in planning safer transportation networks	10

UNIT-V			
8	a	Classify different types of foundations and discuss their suitability for various soil conditions	8
	b	Explain how drones and 3D printing are transforming modern construction practices in terms of efficiency and sustainability	10
9	a	During monsoon seasons, your town faces recurrent floods affecting roads and buildings. Analyze how modern tools such as drones, Artificial Intelligence (AI), and sensors can be used to reduce flood risks	10
	b	Discuss the applications of Artificial Intelligence in civil engineering	8

Signature of Scrutinizer	Signature of Chairman
Name –Dr Madhavi K Dr Sindhu D	Name- Dr Anjaneyappa

RV COLLEGE OF ENGINEERING®
Autonomous Institution affiliated to VTU
I Semester B. E. DEC-JAN 2026 Examinations
DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT
HS114TC/HS124TC-FUNDAMENTALS OF INDIAN CONSTITUTION
(2025 SCHEME)

Time: 03 Hours

Maximum Marks: 50

Instructions to candidates:

Note: Answer all the Questions.

PART – A (Objective type for one marks)
 (True & false and match the following questions are not permitted)

PART-A

1.	Which one of the following has been wrongly listed as a special feature of Fundamental Rights in India? A. Fundamental Rights are more sacrosanct than rights granted by ordinary laws B. Fundamental Rights are subject to reasonable restrictions C. Fundamental Rights are justiciable and can be enforced through the Supreme Court D. None of these	01	CO1
2.	The Sikhs in India are permitted to carry Kirpan. Under which one of the following Fundamental Rights are they permitted to do so? A. Right to freedom B. Right to freedom of religion C. Right to life and liberty D. None of the above.	01	CO1
3.	The main objective of the cultural and educational rights granted to the citizens is A. To preserve the rich cultural heritage of India B. To evolve a single integrated India culture C. To help the minorities to conserve their culture D. All the above	01	CO1
4.	Legal equality under the Indian Constitution implies that (A) Everybody is equal before law (B) There should be equality amongst equals and inequality among unequals (C) There should be no special law for any category of people (D) The state cannot enact different laws for different groups of people	01	CO1
5.	The Indian parliamentary system is different from the British parliamentary system in that India has (A) Both a real and a nominal executive (B) A system of collective responsibility (C) Bicameral Legislature (D) The system of judicial review	01	CO1
6.	Fundamental Rights' are (A) Justifiable (B) Non-justifiable (C) Flexible (D) Rigid	01	CO1
7.	Which of the following is given the power to enforce the Fundamental Rights by the Constitution? (A) All Courts in India (B) The Parliament	01	CO1
8.	Which one of the following statements is correct? (A) Rights are claims of the State against the citizens (B) Rights are privileges which are incorporated in the Constitution of a state (C) Rights are claims of the citizens against the State (D) Rights are privileges of a few citizens against the many	01	CO1

9.	In the context of India, which one of the following is the correct relationship between Rights and Duties? (A) Rights are correlative with Duties (B) Rights are personal and hence independent of society and Duties (C) Rights, not Duties, are important for the advancement of the personality of the citizen (D) Duties, not Rights, are important for the stability of the State	01	CO1
10.	Fundamental Rights (A) Cannot be suspended (B) Can be suspended by order of Prime Minister (C) May be suspended on the will of President (D) May be suspended during Emergency	01	CO1
11.	In the Indian Constitution of the 'Right to Freedom' is granted by four Articles which are (A) Article 19 to 22 (B) Article 16 to 19 Article (C) 17 to 20 (D) Article 18 to 21	01	CO1
12.	Which one of the following is not included in the fundamental right to equality as enshrined in the Indian Constitution? (A) Equality before law (B) Social equality (C) Equal opportunity (D) Economic equality	01	CO1
13.	Which one of the following right of Indian Constitution guarantees all the fundamental rights to every resident of a country? (A) Right against exploitation (B) Right to freedom (C) Right to equality (D) Right to constitutional remedies	01	CO1
14.	Which one of the following is true with respect to the Fundamental Rights of Indian Constitution? (A) The sovereignty of the people (B) Equality of opportunity for all resident (C) Limited government (D) Democracy	01	CO1
15.	B. R. Ambedkar termed Article 32 of the Indian Constitution as the "Heart and Soul of the Indian Constitution". Which one of the following fundamental right it contains? (A) Right to freedom (B) Right to constitutional remedies (C) Right to elementary education (D) Right to freedom of religion	01	CO1
16.	Which of the following Articles contain the right to religious freedom? (A) 25 – 28 (B) 29 – 30 (C) 32 – 35 (D) 23 – 24	01	CO1
17.	Which of the following can a court issue for enforcement of Fundamental Rights? (A) A decree (B) A writ (C) An Ordinance (D) A notification	01	CO1
18.	Which of the following is correct with respect to "Right against exploitation"? (A) Prohibition of traffic in human beings and forced labour (B) Freedom as to payment of taxes for the promotion of any particular religion (C) Protection of interests of minorities	01	CO1

	(D) Equality before law		
19.	In which part of the Indian Constitution, the Fundamental rights are provided? (A) Part II (B) Part III (C) Part V (D) Part IV	01	CO1
20.	Prohibition of discrimination on grounds of religion, race, caste, sex or place of birth is a fundamental right classifiable under? (A) Right to freedom of religion (B) Right to equality (C) Right against exploitation (D) None of these	01	CO1
21.	The Constitution grants Cultural and Educational Rights with a view to (A) Evolve a common national culture. (B) Eradicate illiteracy from the country. (C) To help the minorities conserve their culture (D) None of the above.	01	CO1
22.	Right to Cultural and Educational Right implies (A) Provision of free and compulsory education up to High School (B) Special assistance to backward classes to educate them (C) Provision of religious and cultural instructions in schools run by the government or receiving aid out of government funds (D) Right of minorities to establish and administer educational institutions of their choice.	01	CO1
23.	Which one of the following writs is issued by the court in case of an illegal detention of a person? (A) Habeas Corpus. (B) Mandamus (C) Certiorari. (D) Quo Warranto	01	CO1
24.	Which one of the following writs is issued by the High Court to lower court or a nonjudicial public institution to stop proceedings in a particular case? (A) Habeas Corpus. (B) Prohibition. (C) Quo Warranto (D) Certiorari.	01	CO1
25.	Under which one of the following writs a person can be prevented from taking an action to which he is officially not entitled? (A) Mandamus. (B) Quo Warranto. (C) Certiorari. (D) Mandamus.	01	CO1
26.	To which of the following rights the foreigners are entitled under the Indian Constitution (A) Right against exploitation (B) Right to equality before law. (C) Right to life and Personal Liberty (D) All the above rights	01	CO1
27.	What are DPSPs in the Indian Constitution? A) Fundamental Rights guaranteed to citizens B) Directives given to the President by the Prime Minister C) Guidelines for the governance of the country D) Directives given to the Supreme Court by the Parliament	01	CO2

28.	DPSPs primarily aim to promote the welfare of which group? A) Women B) Children C) Weaker sections of society D) All	01	CO2
29.	DPSPs are primarily the responsibility of which level of government in India? A) Central government B) State governments C) Local governments D) All	01	CO2
30.	The Directive Principles of State Policy in the Indian Constitution are based on the philosophy of which Indian leader? A) Jawaharlal Nehru B) Sardar Vallabhbhai Patel C) Dr. B.R. Ambedkar D) Dr. Rajendra Prasad	01	CO2
31.	According to DPSPs, what should the State do to promote the educational and economic interests of weaker sections of society, especially Scheduled Castes and Scheduled Tribes? A) Provide reservation in public employment and education B) Provide tax benefits to multinational corporations C) Establish state-funded religious institutions D) Ban child labour completely	01	CO2
32.	Who administers the oath of office to the President of India? A) Chief Justice of India B) Speaker C) Vice President D) Prime Minister	01	CO2
33.	Which Constitutional Amendment Act was passed to provide compulsory education to children between 6 and 14 years? A) 82nd B) 83rd C) 86th D) 84th	01	CO2
34.	Which among the following statements is/are true about the Money Bill? 1. Money Bill can be introduced only in the Lok Sabha 2. The Speaker of the Lok Sabha carries the final decision-making authority on whether a Bill is a Money Bill or not 3. Rajya Sabha can only make recommendations to the Money Bill, it cannot change it or reject it 4. Money Bill can be introduced only after the recommendation of the President A) Only 1, 2 and 3 B) Only 2 and 4 C) Only 2 D) All	01	CO2
35.	The President nominates 12 members of the Rajya Sabha according to A) Their performance as office bearers of cultural societies. B) The recommendations made by the Vice President. C) Their role played in political set up of the country. D) Their distinction or specialization in the field of science, art, literature and social service.	01	CO2
36.	Election to the house of the people and the Legislative Assemblies of States in India are conducted on the basis of A) Single Transferable vote B) Proportional Representation C) Limited suffrage D) Adult Franchise	01	CO2

37.	Which of the following are not included in the original jurisdiction of the Supreme Court? A) A dispute between the Government of India and one or more States. B) Violation of Fundamental Rights C) A dispute between the Government of India and other country . D) A dispute between two or more States.	01	CO2
38.	Which of the following categories of disputes is excluded from the original jurisdiction of the Supreme Court? A) Between the Government of India and one or more States B) Between the Government of India and one or more states on one side and one or more other States on the other C) Between two or more States inter se D) Between residents of two or more states	01	CO2
39.	Who decides on the disqualification of the members of the legislative assembly? A) Speaker of legislative assembly B) Governor C) Chief Minister D) None	01	CO2
40.	The procedure for the election of the President of India can be modified through an amendment in the Constitution which must be passed by (A) two-thirds majority by the Lok Sabha (B) two-thirds majority by the Lok Sabha as well as the Rajya Sabha (C) two-thirds majority by the Rajya Sabha (D) two-thirds majority by both Lok Sabha and Rajya Sabha and be ratified by legislatures of atleast half of the states	01	CO2
41.	Special provisions can be made by government for (A) women & children (B) Backward classes (C) SC & ST (D) All	01	CO3
42.	Which of the following is not the basis for the proclamation of “National Emergency”? A) War B) External Aggression C) Armed rebellion D) Internal disturbance	01	CO3
43.	The Governor of a State (1) Is appointed by the President. (2) Holds office during the pleasure of the President. (3) Is the head of the State’s Executive Power. (4) Normally holds office for five years. Select the correct answer from the codes given below. A) 1 and 2 B) 1, 2 and 3 C) 1, 2 and 4 D) All	01	CO3
44.	Which is the Upper Chamber of State Legislature in India? A) Legislative Council B) Legislative Assembly C) Governor’s Office D) None of the above	01	CO3
45.	Which of the following is not the work of Election Commission? (A) To conduct the election of Lok Sabha, Legislative Assembly, President and Vice	01	CO3

	President (B) Approval of setting up of political parties (C) Deciding the disqualification of a Member of Parliament by using his discretionary power. (D) Advising the governor on matters related to the disqualification of members of the Legislative Council		
46.	Governor of a State does not appoint (A) Chief Minister (B) Members of State Public Service Commission (C) Judges of High Court (D) Advocate General	01	CO3
47.	What is the primary function of a state legislature? A) Enforcing laws B) Enacting the law C) Interpreting laws D) Adjudicating disputes	01	CO3
48.	The Governor can make laws through ordinances. (a) During the recess of the state legislature (b) On certain subjects even when Parliament is not in session (c) Only on subjects contained in the concurrent list (d) Under no circumstances	01	CO3
49.	Which of the following statements is correct? (A) The powers of the Chief Election Commissioner and two other Election Commissioners vary in many ways. (B) The salary and allowances of the Election Commissioner are similar to that of the High Court Judge. (C) The Governor appoints State Election Commissions on the advice of the Chief Minister. (D) The term of the Election Commissioners is up to 6 years or 65 years of age.	01	CO3
50.	In which of the following amendment the term of Lok Sabha increased from 5 to 6 years? A) 40th Amendment B) 42nd Amendment C) 44th Amendment D) 46th Amendment	01	CO3

USN

--	--	--	--	--	--	--	--	--	--

RV COLLEGE OF ENGINEERING®
An Autonomous Institution Affiliated to VTU
I Semester B. E. January 2026 Examinations
Computer Science Engineering
BASICS OF JAVA PROGRAMMING

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 is compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.

		PART-A	M	BT	CO
1	1.1	Name two floating-point data types in Java.	1	1	2
	1.2	Which package contains basic Java classes?	1	1	1
	1.3	What are the components of a class in Java?	1	1	1
	1.4	Is String a primitive data type?	1	2	1
	1.5	Give an example of accessing parent method using super.	1	2	3
	1.6	Can Java support multiple inheritance using classes?	1	2	3
	1.7	What access modifier are methods in an interface by default?	1	1	1
	1.8	Write the syntax to implement an interface.	1	2	4
	1.9	Justify: Implementing Runnable is better than extending Thread.	1	2	4
	1.10	List any two methods of the Thread class.	1	1	2

PART B (Maximum subdivisions is limited to 3 in each questions)

UNIT I				
2	a	Explain the principles of OOP in Java with a program showing encapsulation and inheritance.	18	
	b	Write a Java program that uses a looping statement and a switch statement to perform basic arithmetic operations. The program should repeatedly display a menu for addition, subtraction, multiplication, and division using a do-while loop, take the user's choice using a switch statement, perform the selected operation on two numbers, and continue until the user chooses to exit.		

UNIT II			
3	a	What is a class in Java? Explain its structure.	18
	b	Create a method to calculate the factorial of a number.	

UNIT III			
4	a	Define inheritance. Explain the different types of inheritance in Java.	18
	b	Write a Java program to demonstrate method overriding using the classes Animal, Dog, and Cat.	

OR

5	a	Create an abstract class named Shape with an abstract method area(). Derive two subclasses, Circle and Rectangle, that implement the area() method to calculate and display their respective areas. Demonstrate the use of runtime polymorphism in the main() method.	18
	b	What is the purpose of the super keyword in Java? Write a program using super() to call parent constructor.	

UNIT IV			
6	a	Create an integer array and attempt to access an invalid index inside a try block. Catch the exception using a catch block and display a suitable message.	18
	b	Write a Java interface named Vehicle and implement it in Car and Bike classes.	

OR

7	a	Create a package named shapes containing a class Circle with a method area() that calculates and displays the area of a circle. Create another package named mainapp containing a class AreaDemo that imports the Circle class from the shapes package and calls its method to display the result.	18
	b	What are checked and unchecked exceptions? Explain with examples.	

UNIT V			
8	a	Describe two ways of creating a thread in Java with examples.	18
	b	Explain the Java thread model with a diagram.	

OR

9	a	What is thread priority? How is it set and retrieved in Java?	18
---	---	---	----

	b	Write a Java program that creates a child thread from the main thread and explain the output. Note: The child thread should display a message five times with a short delay. The main thread should also display its own message so that both run concurrently.	
--	---	---	--

Q.P.Code

Course Code: **CS125BIC**

Course Title: Basics of Java Programming

PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	1	1	1	2	2	2	1	2	2	1
COs	2	1	1	1	3	3	1	4	4	2

PART-B

Question No		B T Levels	Cos addressed	Question No		BT Levels	Cos addressed
2	a	1	1	3	a	1	1
	b	2	2		b	2	3
	c				c		
	d				d		
4	a	1	1	5	a	3	4
	b	2	3		b	1	2
	c				c		
	d				d		
6	a	3	4	7	a	3	4
	b	2	3		b	1	1
	c				c		
	d				d		
8	a	1	2	9	a	1	2
	b	1	1		b	2	3
	c				c		
	d				d		
10	a			11	a		
	b				b		
	c				c		
	d				d		

Signature of Scrutinizer:

Signature of Chairman

Name:

Name:

RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
I Semester B. E. January-2026 Examinations
Department of Mathematics
FUNDAMENTALS OF LINEAR ALGEBRA, CALCULUS AND STATISTICS
(2025 SCHEME)

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 and 3 are compulsory. Answer any one full question from 4 or 5, 6 or 7 and 8 or 9.
3. Marks will not be awarded for direct answers without supporting steps.
4. Write answers up to four decimal places.
5. Use of first year Mathematics Handbook for B.E. is permitted. Do not write anything on handbook.

Part - A			
1	1.1	If k is an irrational number, then rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & k \\ 0 & 2 & k \end{bmatrix}$ is ____.	01
	1.2	If $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -2 & 1 & 2 \end{bmatrix}$, then the dominant eigenvalue of A^{-1} is ____.	01
	1.3	Deduce the polar equation of the circle having radius a centered at $(a, 0)$.	02
	1.4	If $y^x = \sin x$, determine $\frac{dy}{dx}$ using partial differentiation.	02
	1.5	Sketch the region of integration for the integral $\int_1^4 \int_1^{\sqrt{y}} f(x, y) dx dy$.	02
	1.6	Determine mean of x and y if two regression equations of the variables x and y are $x = 19.13 - 0.87y$, $y = 11.6 - 0.5x$.	02

Part - B

Unit-I			
2	a	Check for the consistency of the following system and solve if consistent $5x + 3y + 7z = 4$; $3x + 26y + 2z = 9$; $7x + 2y + 10z = 5$.	06
	b	Determine an approximate solution of the following system of linear equation by Gauss-Siedel method with initial approximation $[1 \ 0 \ 0]^T$. Perform 4 iterations. $8x + y - z = 8$; $2x + y + 9z = 12$; $x - 7y + 2z = -4$.	06
	c	Determine the largest eigenvalue and the corresponding eigenvector of the matrix	06

		$A = \begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix}$. Choose the initial vector as $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Perform 3 iterations.	
--	--	---	--

Unit-III			
3	a	Determine the angle of intersection of the lemniscate $r^2 = a^2 \cos 2\theta$ and the cardioid $r = a(1 + \cos \theta)$.	06
	b	Find the coordinates of the centre of curvature at $(a, 2a)$ on the parabola $y^2 = 4ax$.	06
	c	Obtain the expansion of the function $e^{\sin(x)}$ in ascending powers of x up to terms containing x^4 .	06

Unit-III			
4	a	Jacobian represents transformation function to traverse between coordinate system. If $x = u \cos v$ and $y = u \sin v$, determine $\frac{\partial(u,v)}{\partial(x,y)}$.	05
	b	If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, prove that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.	05
	c	The temperature T at any point (x, y, z) in space is $T = xyz^2$. Determine the highest temperature on the surface of the sphere $x^2 + y^2 + z^2 = a^2$, using Lagrange Method.	08
OR			
5	a	The steady state temperature of a circular metal plate is $u = r^2 \cos(2\theta) - r \cos(\theta) + 2$. Show that u satisfies the Laplace equation $u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0$.	06
	b	If $u = g(2x - 3y, 3y - 4z, 4z - 2x)$, prove that $6 \frac{\partial u}{\partial x} + 4 \frac{\partial u}{\partial y} + 3 \frac{\partial u}{\partial z} = 0$.	06
	c	The lengths x, y and z of edges of a rectangular box are changing with time. At the instant in question $x = 1 \text{ m}$, $y = 2 \text{ m}$, $z = 3 \text{ m}$ $dx/dt = dy/dt = 1 \text{ m/sec}$ and $dz/dt = -3 \text{ m/sec}$. At what rates the box's volume, surface area and length of diagonal are changing at that instant?	06

Unit-IV			
6	a	Determine the area of the region inside the cardioid $r = 4(1 + \cos \theta)$ and outside the circle $r = 4$, using double integrals.	09
	b	A metal plate is in the form of a triangle with vertices $(0,0)$, $(1,0)$ and $(1,1)$. The density at any point on the metal plate is given by $\rho(x,y) = xy$. Determine the center of gravity of the metal plate.	09
OR			
7	a	Using triple integral obtain the volume of the solid bounded between the cone $z^2 = x^2 + y^2$ and the plane $z = 2$.	09
	b	Evaluate $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$ by changing the order of integration.	09

Unit-V			
--------	--	--	--

8	a	Compute the correlation coefficient for this data. Also find the regression lines of y on x and x on y . <table><tr><td>x</td><td>5</td><td>7</td><td>8</td><td>10</td><td>11</td><td>13</td><td>16</td></tr><tr><td>y</td><td>33</td><td>30</td><td>28</td><td>20</td><td>18</td><td>16</td><td>9</td></tr></table>	x	5	7	8	10	11	13	16	y	33	30	28	20	18	16	9	09
x	5	7	8	10	11	13	16												
y	33	30	28	20	18	16	9												
	b	Fit a second-degree polynomial $y = a + bx + cx^2$ for the following data using least square method. <table><tr><td>x</td><td>-3</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>y</td><td>4.63</td><td>2.11</td><td>0.67</td><td>0.09</td><td>0.63</td><td>2.15</td><td>4.58</td></tr></table>	x	-3	-2	-1	0	1	2	3	y	4.63	2.11	0.67	0.09	0.63	2.15	4.58	09
x	-3	-2	-1	0	1	2	3												
y	4.63	2.11	0.67	0.09	0.63	2.15	4.58												
OR																			
9	a	From the following frequency distribution compute first four moments about $A = 15$ and hence compute first four central moments. <table><tr><td>Size x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>30</td><td>35</td></tr><tr><td>Frequency f</td><td>4</td><td>10</td><td>20</td><td>36</td><td>16</td><td>12</td><td>2</td></tr></table>	Size x	5	10	15	20	25	30	35	Frequency f	4	10	20	36	16	12	2	09
Size x	5	10	15	20	25	30	35												
Frequency f	4	10	20	36	16	12	2												
	b	A biology student is monitoring the growth of a bacterial culture. At successive 2-hour intervals she records the number of colonies shown below. <table><tr><td>Time (x)</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>Colony count (y)</td><td>1</td><td>3</td><td>6</td><td>12</td><td>24</td></tr></table> Fit a curve of the form $y = ab^x$ to the data by the method of least squares.	Time (x)	2	4	6	8	10	Colony count (y)	1	3	6	12	24	09				
Time (x)	2	4	6	8	10														
Colony count (y)	1	3	6	12	24														

USN

--	--	--	--	--	--	--	--	--	--

COURSE CODE: ME113ATE / ME123ATE

RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
I/II Semester B.E. Examinations
DEPARTMENT OF MECHANICAL ENGINEERING
COURSE TITLE: FUNDAMENTALS OF MECHANICAL ENGINEERING
(2025 SCHEME)

Time: 03 Hours Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first two pages of the answer booklet only.
2. Answer FIVE full questions from Part B. In Part B questions 2 and 3 are compulsory.
Answer any one full question from 4 and 5, 6 and 7, 8 and 9.

PART-A

(Part – A, answers should be written in first two pages of the answer booklet only)

			M	CO	BT
1	1.1	Nylons and polyethylene are _____ type of plastic	1	1	2
	1.2	Ductility is the ability of material to be _____	1	1	2
	1.3	Temperature range in Arc welding is _____ °C	1	1	2
	1.4	Name the alloys of Soft Solder	1	1	1
	1.5	_____ converts linear motion of piston to rotary motion of crankshaft.	1	1	1
	1.6	_____ tool is used for smoothening the surface of the drilled holes.	1	1	1
	1.7	_____ are used to provide airtight gap between piston and cylinder	1	1	1
	1.8	_____ adjusts the frequency and amplitude of the alternating current with the help of a microcontroller.	1	1	2
	1.9	Give an example of Closed Loop mechatronic system	1	1	2
	1.10	The polar configuration robots, also called _____ robots	1	1	1

PART-B

UNIT-I			M	CO	BT
2	a	With proper examples differentiate between Thermosetting and Thermoplastic polymers.	5	1	3
	b	With suitable example explain the application of ceramics in the following fields: a) Medical (Bio ceramics) b) Military c) Communications	5	2	1
	c	With a flow chart explain in detail Classification of Engineering materials	8	2	3

UNIT-II			M	CO	BT
3	a	With a neat diagram explain in detail working principle of an Arc welding process. Explain briefly types of electrodes used in Arc welding	8	3	3
	b	In brief explain applications of Welding.	2	2	2
	c	Differentiate between of MIG and TIG welding processes.	8	2	2

UNIT-III			M	CO	BT
4	a	With a neat diagram explain specification of a lathe.	8	2	2
	b	Explain with neat diagram following lathe operations: i) Boring ii) Facing	6	2	2
	c	With a neat diagram explain Tapping drilling operation.	4	1	1
OR					
5	a	Explain with neat diagram following lathe operation: a) Knurling b) Thread Cutting	6	2	1
	b	How Taper turning by Swiveling the compound rest is performed on Lathe? Explain with a neat sketch.	6	3	2
	c	Differentiate between Counter sinking and Counter boring drilling operations with the neat sketches.	6	3	2

UNIT-IV			M	CO	BT
6	a	With P-V diagram explain the working of different strokes of a four stroke Constant Pressure IC engine.	10	3	2
	b	Explain with a neat diagram Parallel-Hybrid vehicle system	8	2	2
		OR			
7	a	With a neat diagram explain Series- Parallel Hybrid electric vehicle	10	3	2
	b	A four cylinder four-stroke engine running at 1000 rpm, develops an indicated power of 15 kW. The mean effective pressure is 5×10^5 N/m ² . Find the diameter of the cylinder and the stroke of the piston when the ratio of diameter to stroke is 0.8.	8	2	2

UNIT-V			M	CO	BT
8	a	Explain following Robot configurations with neat sketches: a) Spherical configuration b) Jointed Arm configuration	10	3	3
	b	With a neat sketch explain in detail Phases of Mechatronics system design process	8	3	2
		OR			
9	a	How Anti-Lock Braking System (ABS) will work? Explain with a block diagram.	6	3	2
	b	Briefly explain Well to Wheel Analysis comparing Petrol scooter vs electric scooter	6	1	1
	c	With a neat sketch explain Elements of a Robotic System	6	2	2